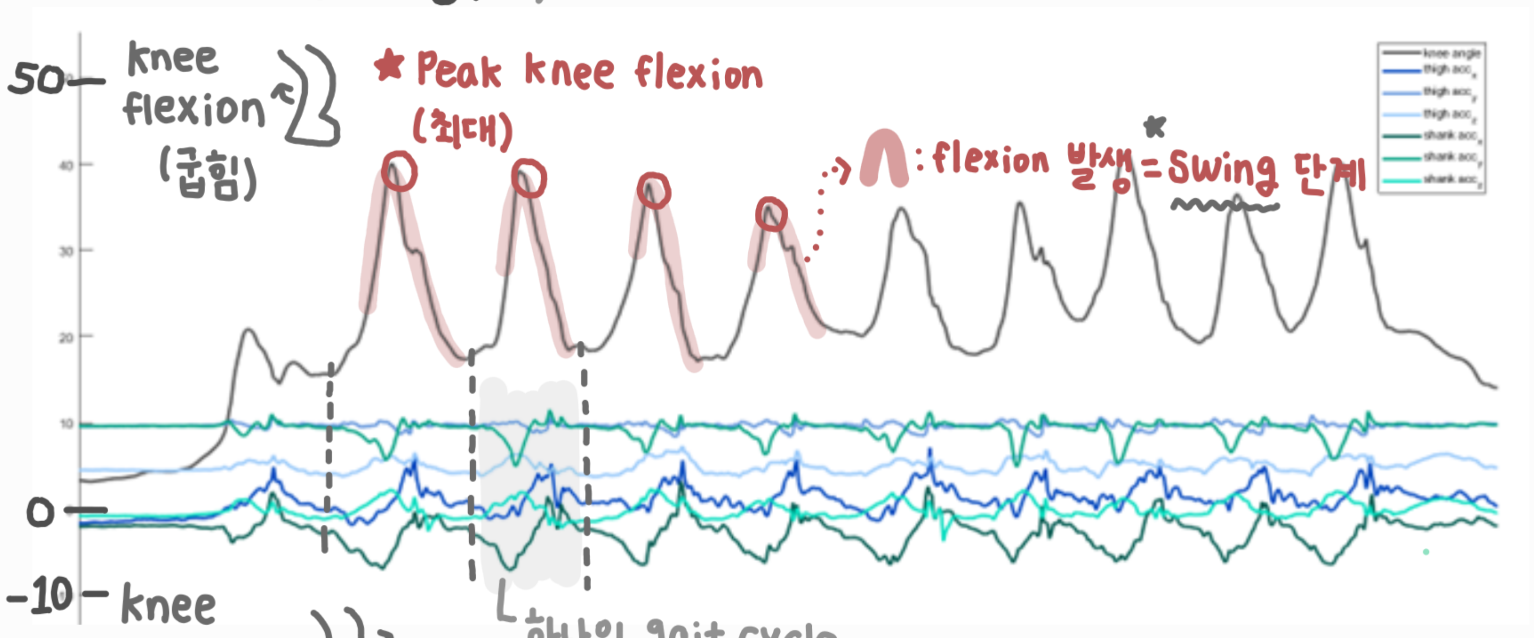
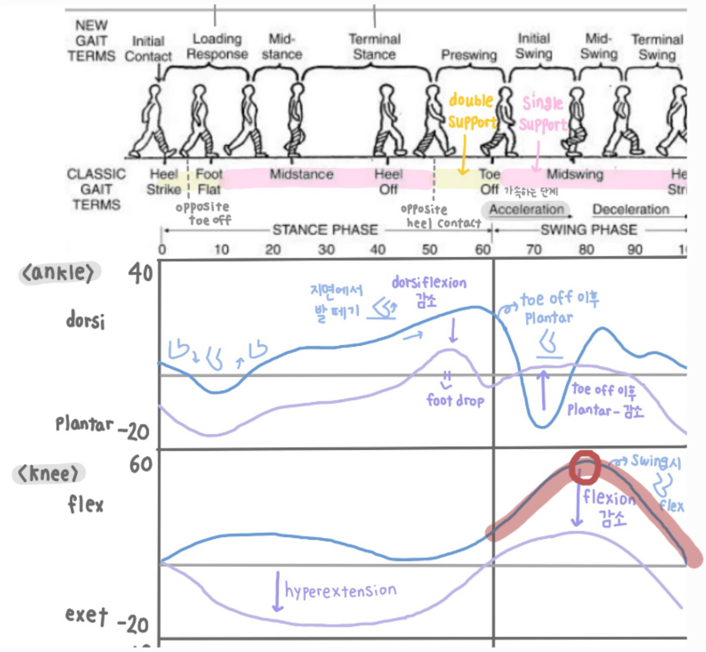
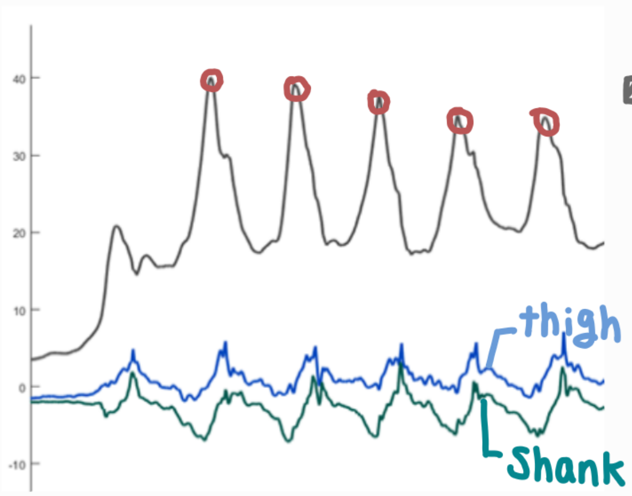
〈LG - 평지 보행〉 1) acc



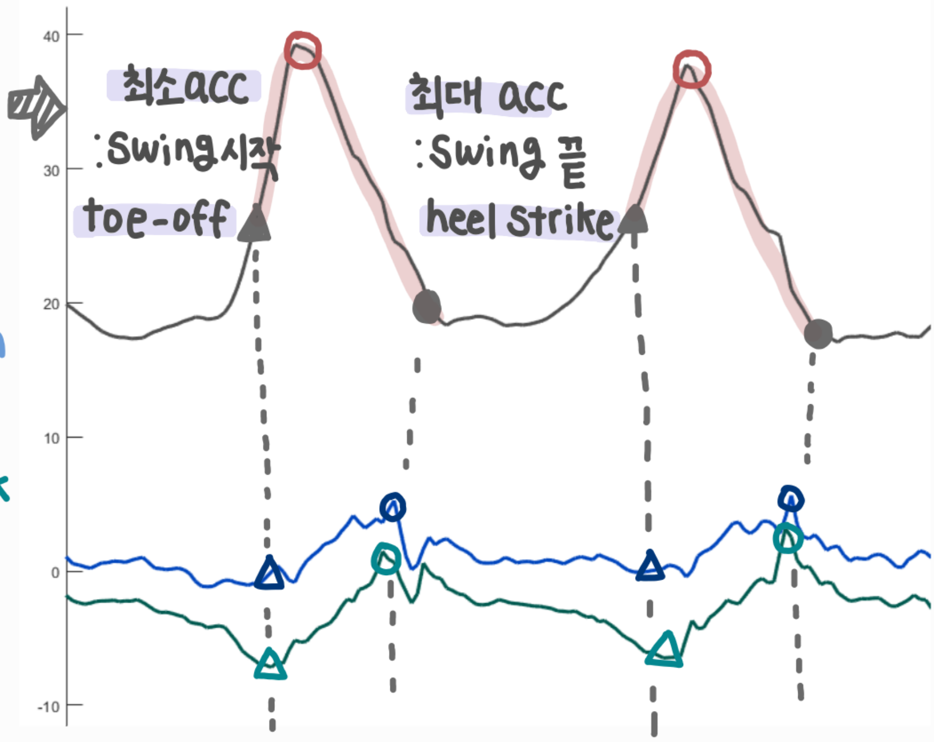


어떠한 data 사용할 것인가?

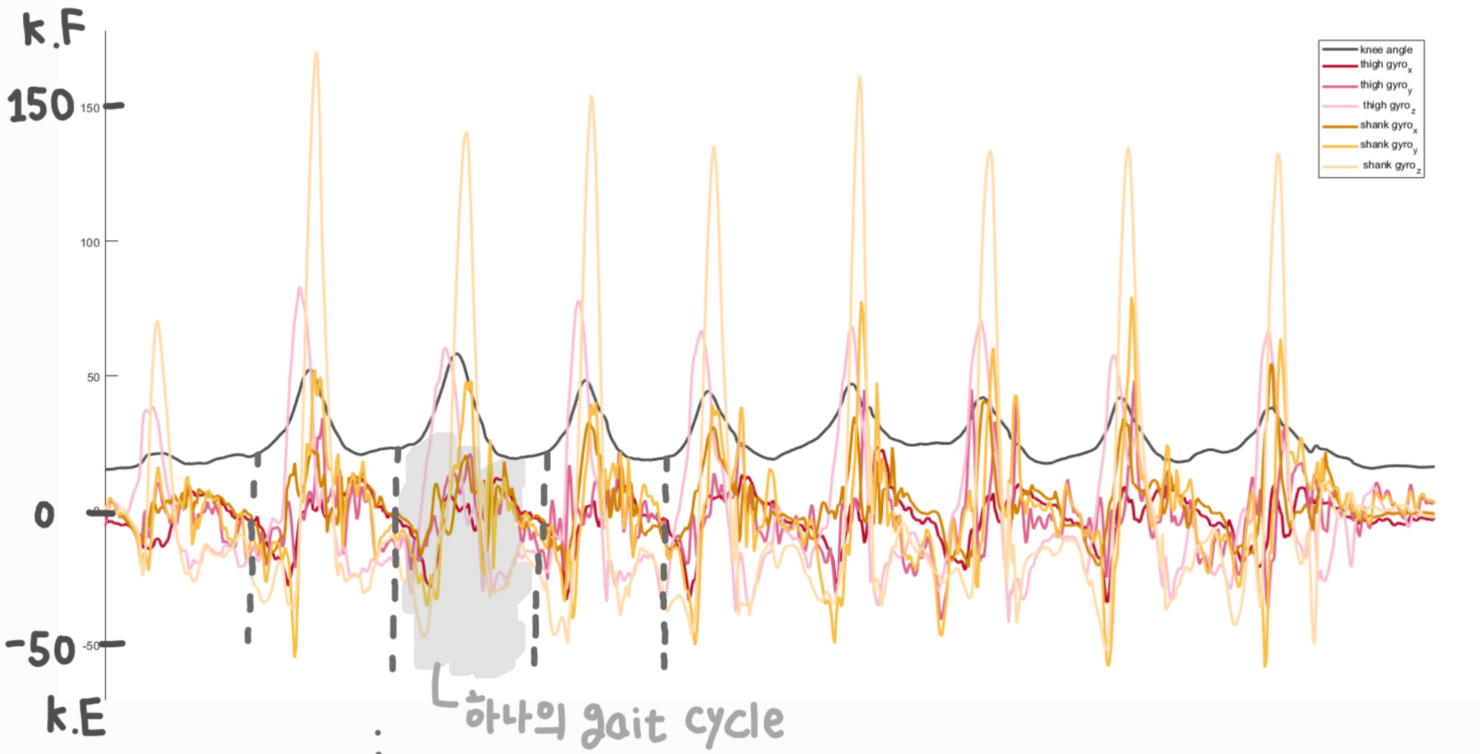
- acc_x : 제일 큰 변동 보임



- acc_x 값만 추출



〈LG - 평지 보행〉 2) gyro



└하나의 gait cycle

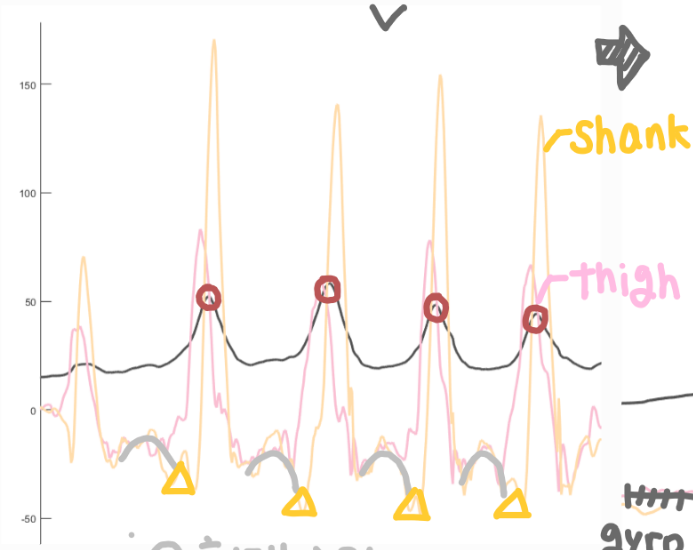
어떠한 data 사용할 것인가?

• gyro_z : 제일 큰 변동 보임

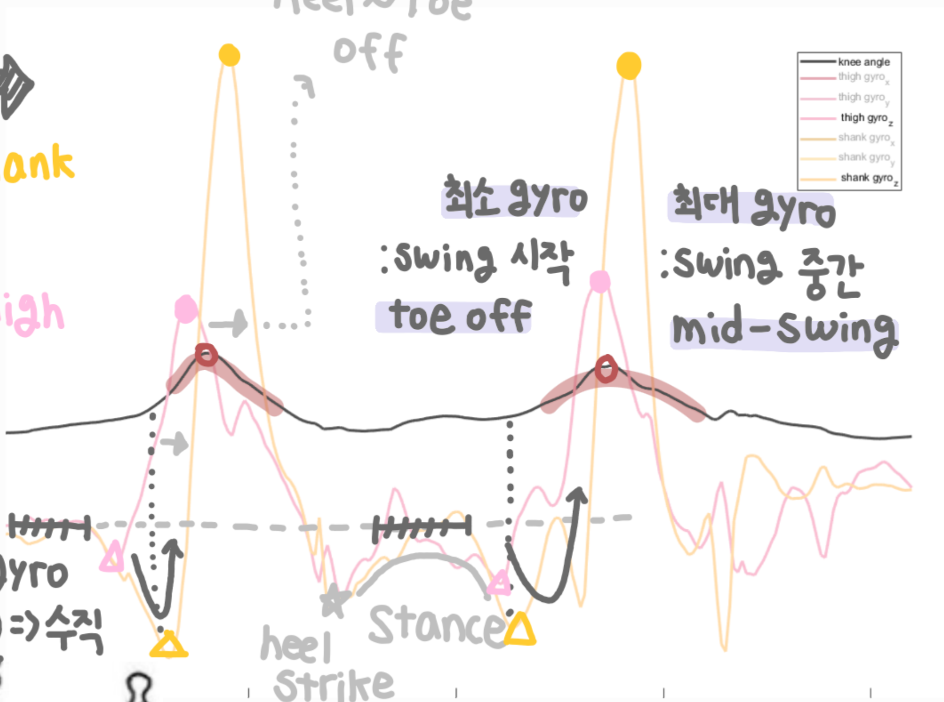
thigh 먼저 이동

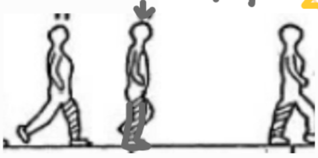
⇒thigh timing 빠름

heel~toe off

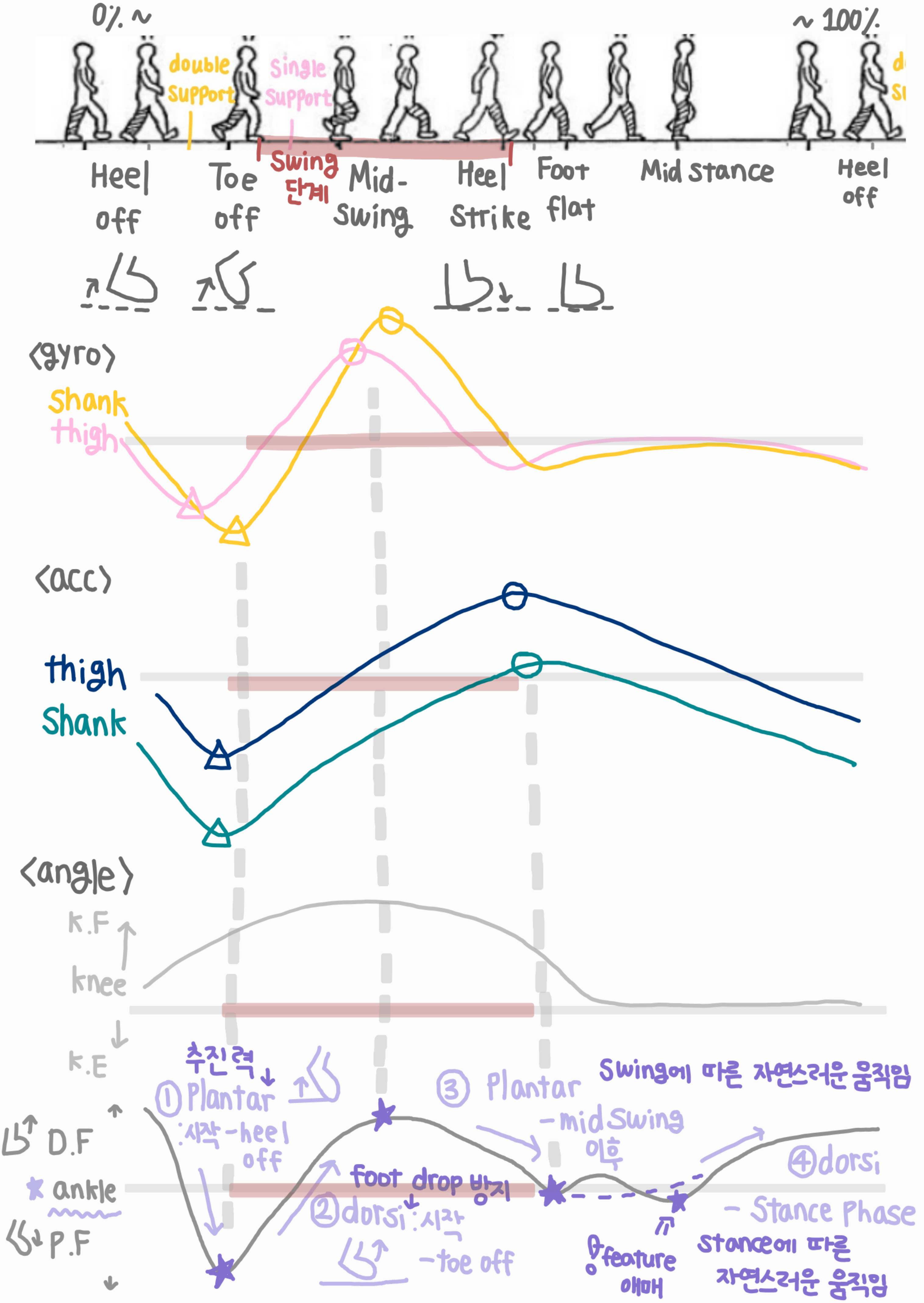


: ∩형태 → stance

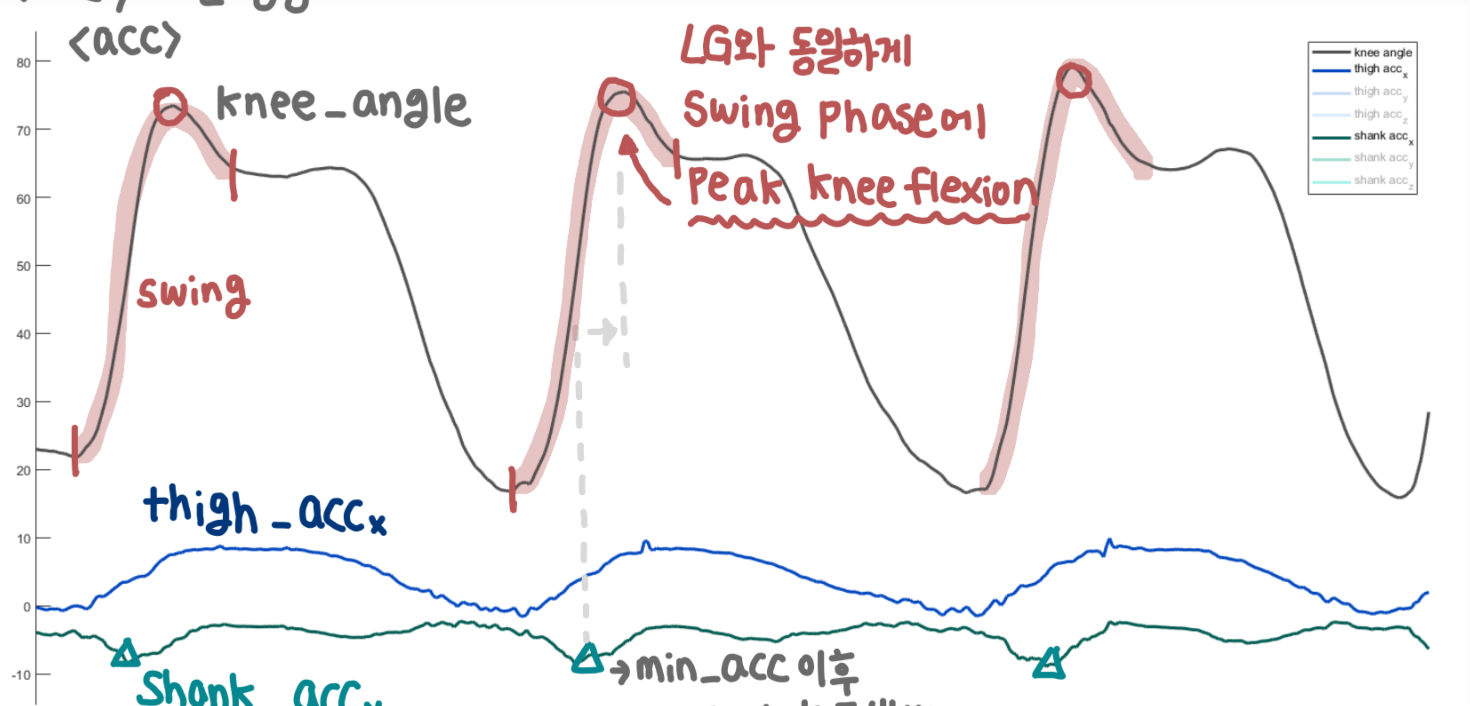




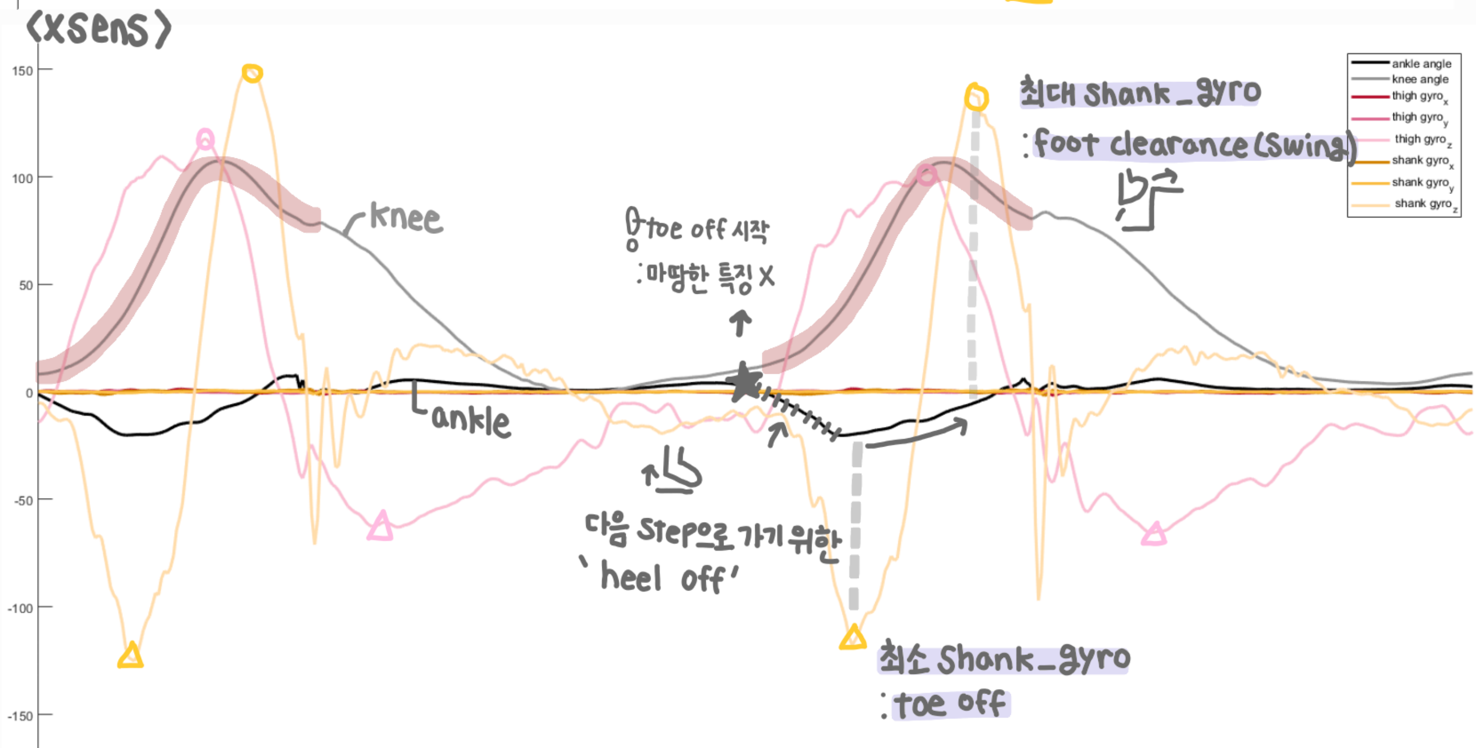
<LG-총정리>


0% ~
~ 100%
double support
single support
Swing 단계
Heel off
Toe off
Mid-swing
Heel Strike
Foot flat
Mid stance
Heel off
<gyro>
Shank
thigh
<acc>
thigh
Shank
<angle>
K.F
knee
K.E
D.F
ankle
P.F
추진력
① Plantar
: 시작 - heel off
foot drop 방지
② dorsi : 시작
- toe off
③ Plantar
- mid swing 이후
Swing에 따른 자연스러운 움직임
④ dorsi
- Stance Phase
feature 애매
Stance에 따른 자연스러운 움직임

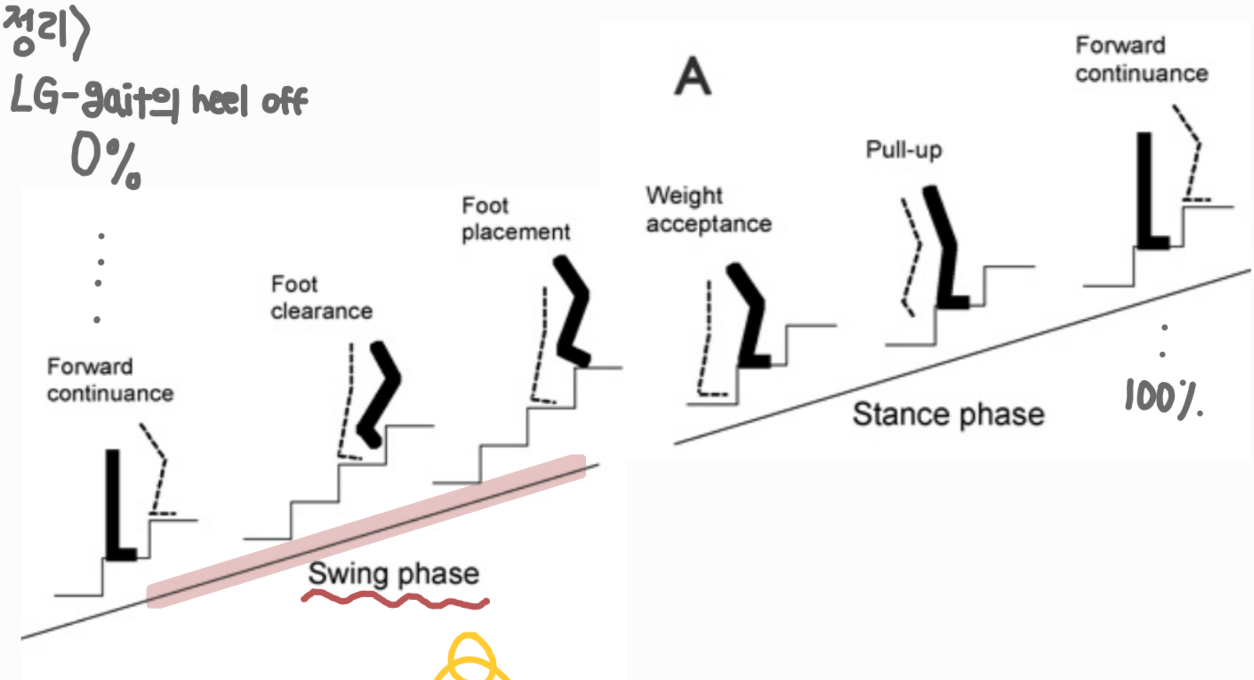
<IC> 계단 상승

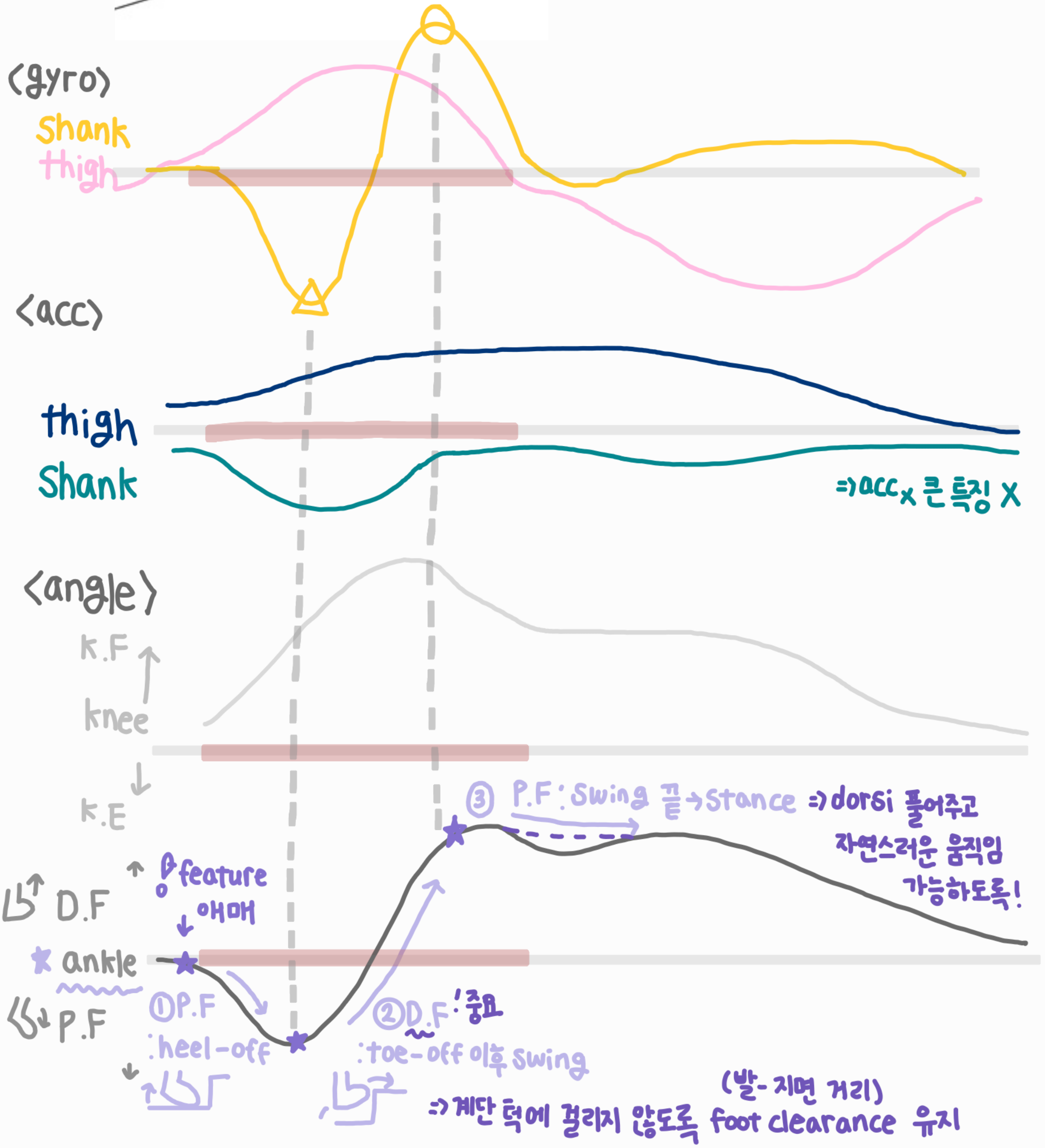

<acc>
knee_angle
swing
LG와 동일하게
Swing Phase에
Peak knee flexion
thigh_accx
Shank_accx
→min_acc 이후
Peak K.F 발생
knee angle
thigh accx
thigh accy
thigh accz
shank accx
shank accy
shank accz


<gyro>
Shank
-gyro z
thigh
-gyroz
knee angle
thigh gyrox
thigh gyroy
thigh gyroz
shank gyrox
shank gyroy
shank gyroz

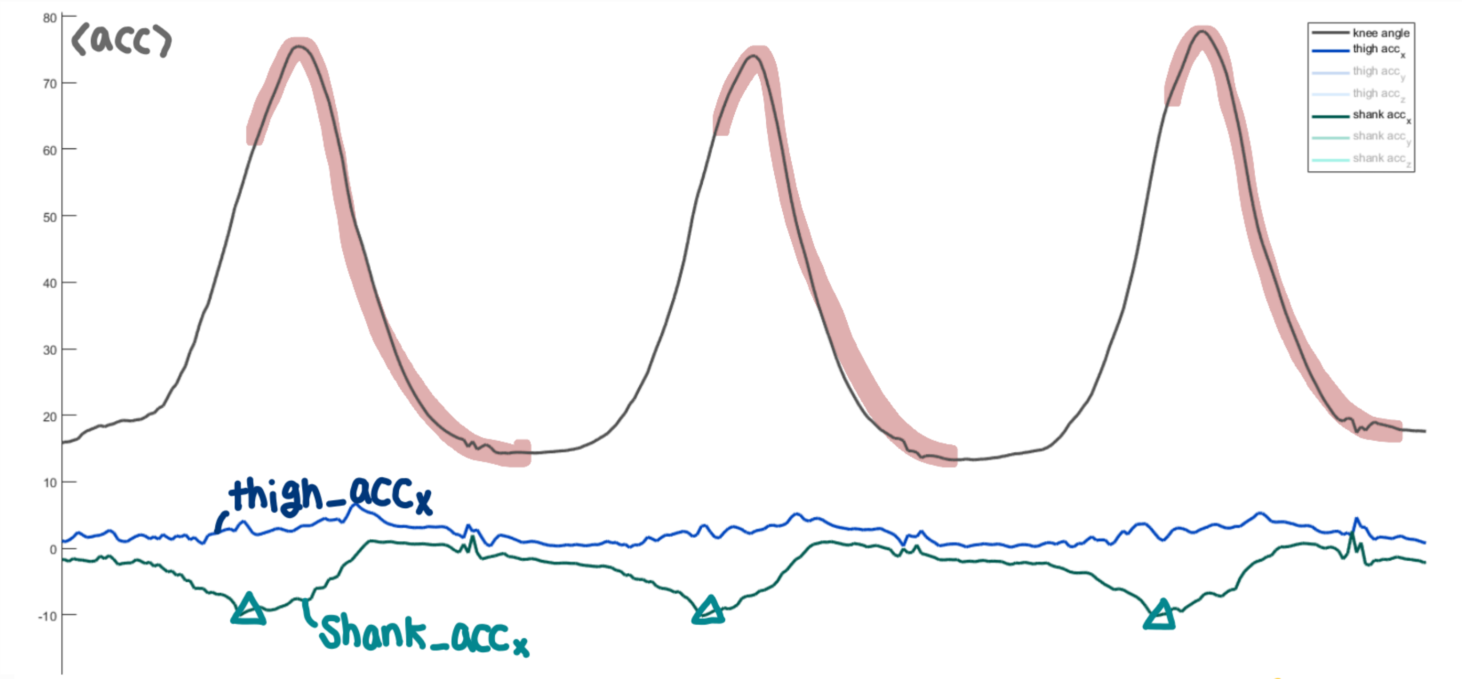

<xsens>
knee
ankle
toe off 시작
:마땅한 특징 X
다음 step으로 가기 위한
'heel off'
최대 shank_gyro
: foot clearance (swing)
최소 Shank_gyro
: toe off
ankle angle
knee angle
thigh gyrox
thigh gyroy
thigh gyroz
shank gyrox
shank gyroy
shank gyroz

<IC 총정리>

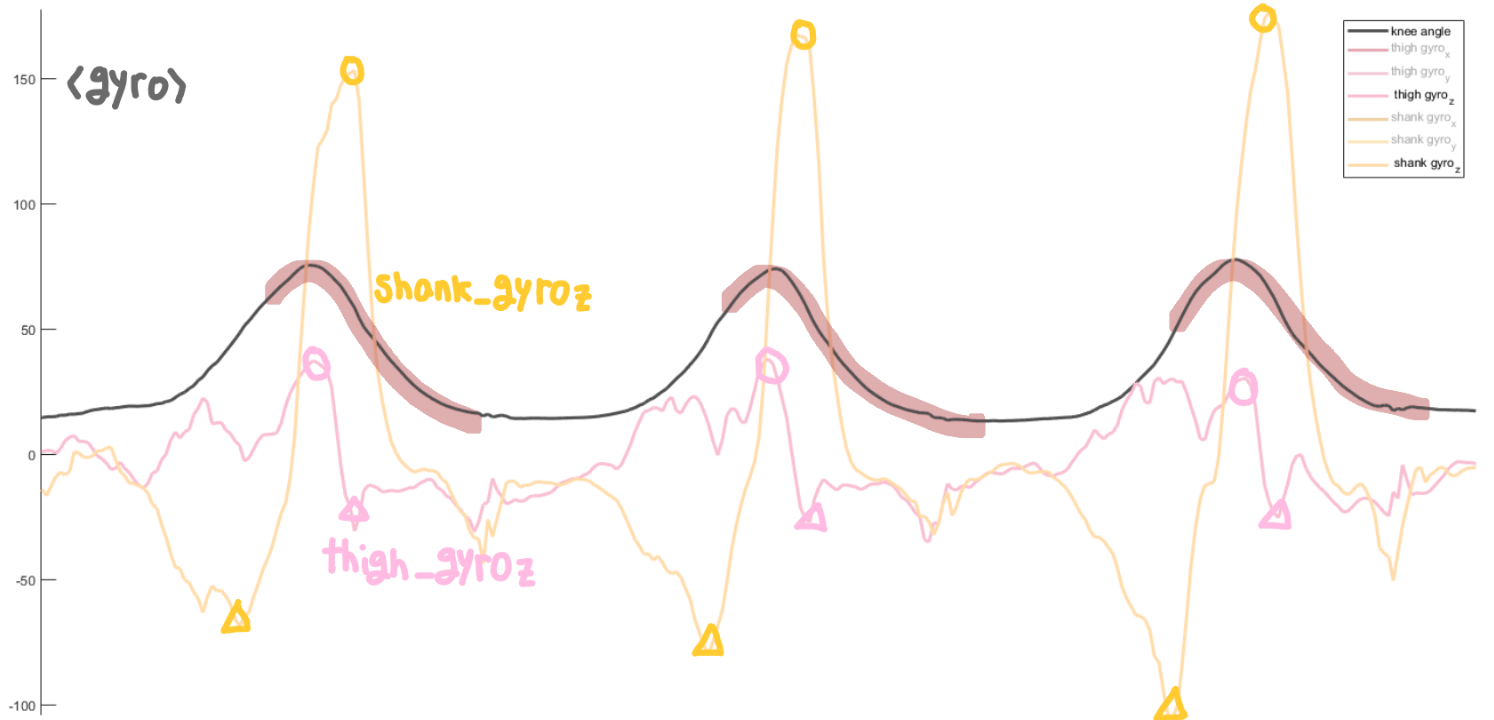

LG-gait의 heel off
0%
A
Forward continuance
Foot clearance
Foot placement
Weight acceptance
Pull-up
Forward continuance
Swing phase
Stance phase
100%.


<gyro>
shank
thigh
<acc>
thigh
Shank
=> acc_x 큰 특징 X
<angle>
K.F
knee
K.E
D.F
ankle
P.F
feature 애매
① P.F : heel-off
② D.F ! 중요 : toe-off 이후 swing
③ P.F : swing 끝 → stance => dorsi 풀어주고 자연스러운 움직임 가능하도록!
(발-지면 거리)
=> 계단 턱에 걸리지 않도록 foot clearance 유지

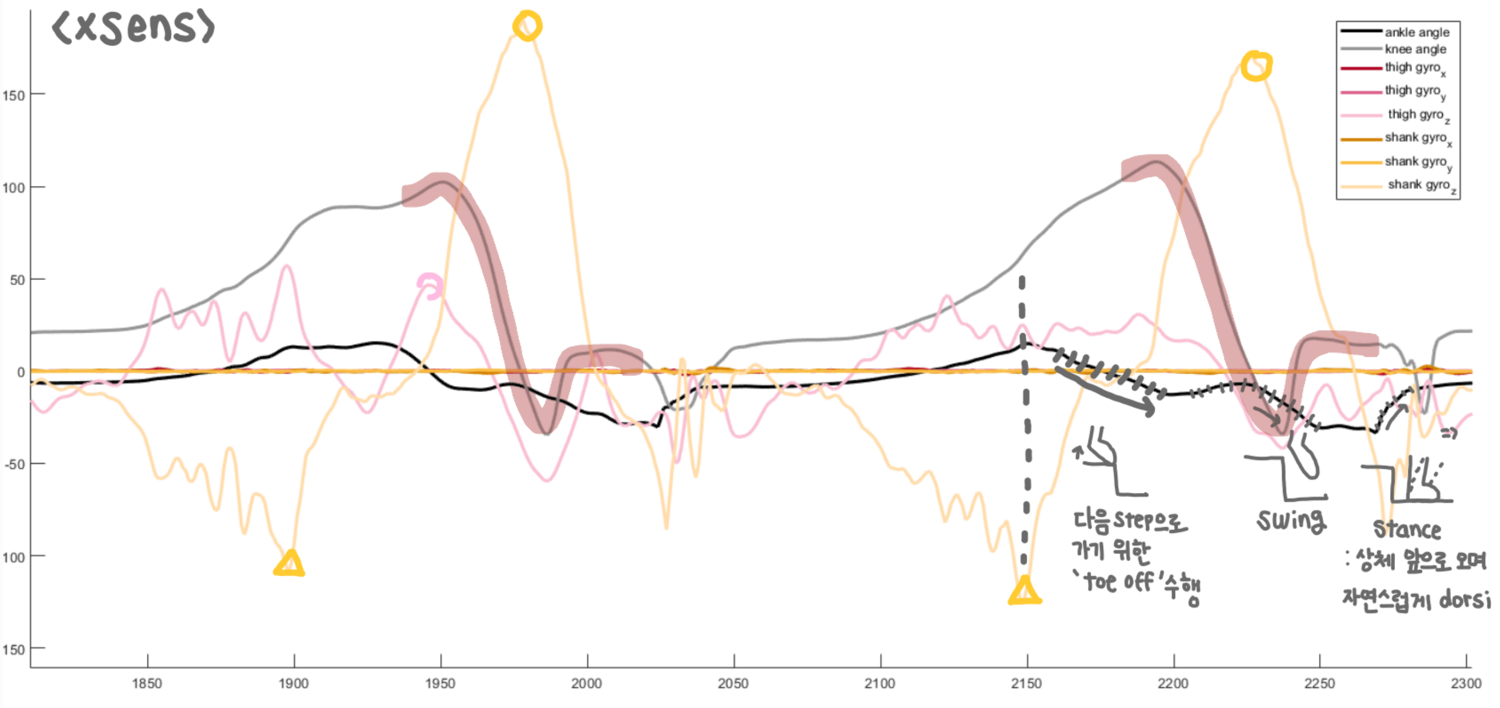
<DC> 계단 하강



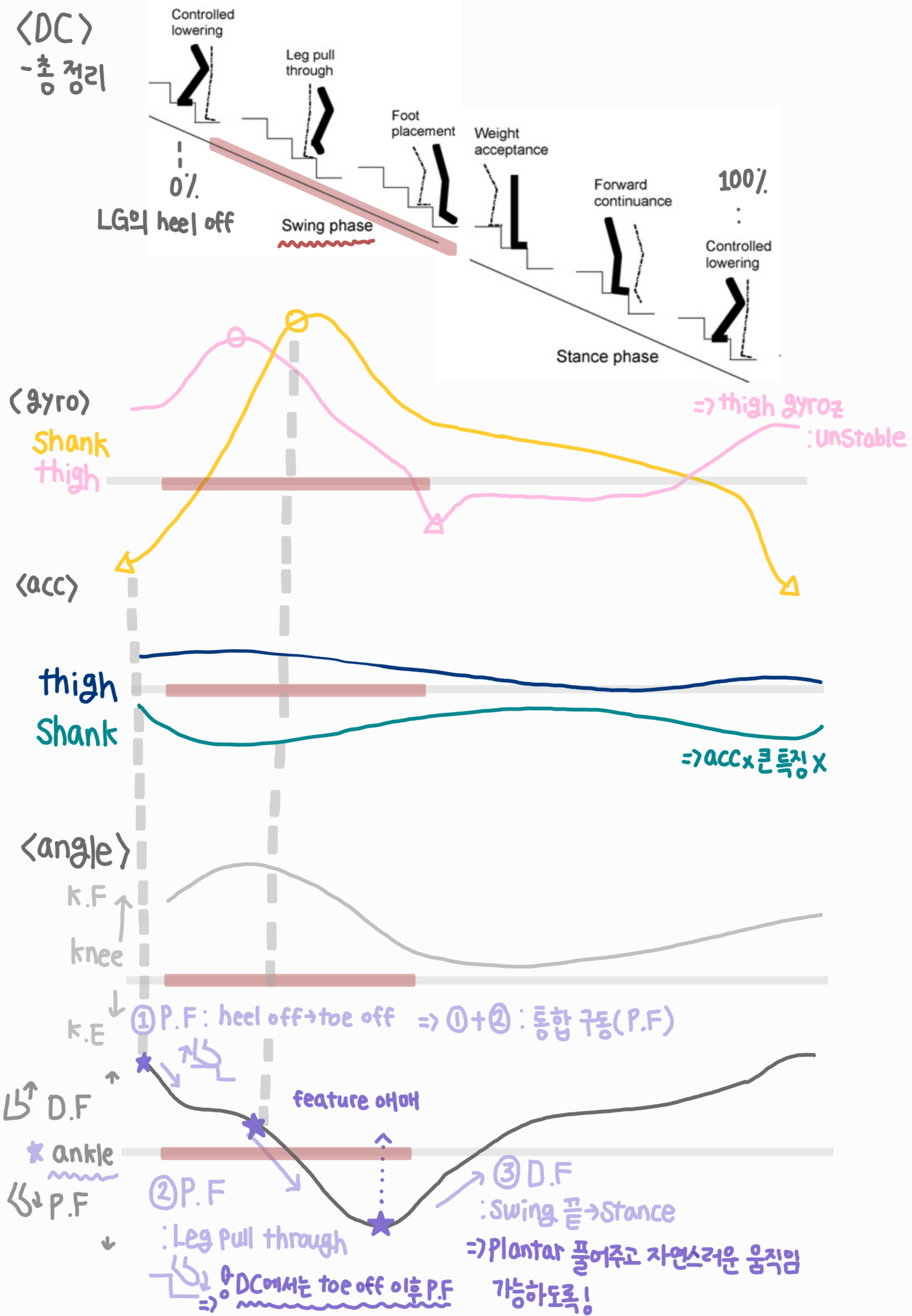
<acc>
knee angle
thigh acc$_x$
thigh acc$_y$
thigh acc$_z$
shank acc$_x$
shank acc$_y$
shank acc$_z$
thigh_acc$_x$
Shank_acc$_x$



<gyro>
knee angle
thigh gyro$_x$
thigh gyro$_y$
thigh gyro$_z$
shank gyro$_x$
shank gyro$_y$
shank gyro$_z$
Shank_gyro$_z$
thigh_gyro$_z$



<xsens>
ankle angle
knee angle
thigh gyro$_x$
thigh gyro$_y$
thigh gyro$_z$
shank gyro$_x$
shank gyro$_y$
shank gyro$_z$
다음 step으로 가기 위한 'toe off' 수행
swing
Stance
: 상체 앞으로 오며 자연스럽게 dorsi



<DC>
-총 정리
Controlled lowering
Leg pull through
Foot placement
Weight acceptance
Forward continuance
Controlled lowering
0%
LG의 heel off
100%
Swing phase
Stance phase
(gyro)
shank
thigh
=> thigh gyroz
: unstable
<acc>
thigh
shank
=> acc x 큰 특징 X
<angle>
K.F
knee
K.E
① P.F : heel off→toe off
=> ①+② : 통합 구동(P.F)
D.F
ankle
P.F
feature 애매
② P.F
: Leg pull through
=> DC에서는 toe off 이후 P.F
③ D.F
: swing 끝→stance
=> plantar 풀어주고 자연스러운 움직임 가능하도록!